time. Then

$$\bar{v}\cos\theta\triangle t = d \ . \qquad (7)$$

Let $\triangle p_{du}$ be the momentum change magnitude. The mean force on the plate is

$$f = \frac{\triangle p_{du}}{\triangle t} = \frac{p_d - p_u}{d/(\bar{v}\cos\theta)} \approx \frac{(p_d - p_u)\, c\cos\theta}{d} \ ; \qquad (8)$$

Substitution gives

$$f \approx 1.04 \times 10^{-9}\ \mathrm{N} \ . \qquad (9)$$

（III） The time for one particle to cross the plate is

$$\triangle t = \frac{d}{\bar{v}\cos\theta} \approx \frac{d}{c\cos\theta} \approx 2.07 \times 10^{-11}\ \mathrm{s} = 0.0207\ \mathrm{ns} \ , \qquad (10)$$

far shorter than pulse period $\tau = 2\mathrm{ns}$. The mean force on the plate during the pulse passage is

$$F \approx (p_d - p_u) j \ ; \qquad (11)$$

Substitution gives

$$F = 0.107\ \mathrm{N} \ . \qquad (12)$$

Using （4）, the particle energy can be written

$$E = \sqrt{p^2c^2 + m^2c^4} \ , \qquad (13)$$

Thus the particle energies before and after crossing are

$$E_d = \sqrt{p_d^2c^2 + m^2c^4} = 63.0\ \mathrm{MeV} \ , \ E_u = \sqrt{p_u^2c^2 + m^2c^4} = 22.8\ \mathrm{MeV} \ . \qquad (14)$$

The heat absorbed by the plate during the pulse is therefore

$$Q = (E_d - E_u) j \tau \ ; \qquad (15)$$

Substitution gives

$$Q = 8.04 \times 10^{-2}\ \mathrm{J} \ . \qquad (16)$$